

GENERAL SPECIFICATION G4.61

Fiberglass Reinforced Plastic and Thermoplastic Tanks

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G4.61 FIBERGLASS REINFORCED PLASTIC AND THERMOPLASTIC TANKS

G4.61.1 General

- (a) This section covers the furnishing of thermoplastic tanks fabricated of fiberglass reinforced plastic (FRP), polyethylene (PE), high density polyethylene (HDPE) and propylene (PP) which are intended for above-ground bulk storage of chemical solutions as indicated herein.
- (b) The following is a list of standards which may be referenced in this section:
 - (i) ASTM International (ASTM):

Standard	Subject
C582	Standard Specification for Contact-Moulded Reinforced Thermosetting Plastic (RTP) Laminates for Corrosion-Resistant Equipment.
D1998 – 13	Standard Specification for Polyethylene Upright Storage Tanks
D2583	Standard Test Method for Indentation Hardness of Rigid Plastics by Means of a Barcol impression tool.
D2584	Standard Test Method for Ignition Loss of Cured Reinforced Resins.
D3299	Standard Specification for Filament-Wound Glass-Fibre-Reinforced Thermoset Resin Corrosion-Resistant Tanks.
D4097	Standard Specification for Contact-Moulded Glass-Fibre-Reinforced Thermoset Resin Corrosion-Resistant Tanks.
E84	Standard Test Method for Surface Burning Characteristics of Building Materials:

- (ii) British Standard Institute:

Standard	Subject
BS EN 13121-1:2003	GRP tanks and vessels for use aboveground. Raw materials. Specification conditions and acceptance conditions
BS EN 13121-2:2003	GRP tanks and vessels for use aboveground. Composite materials. Chemical resistance
BS EN 13121-3:2008 + Amendment 1:2010,	GRP tanks and vessels for use aboveground. Design and workmanship
BS EN 13121-4:2005,	GRP tanks and vessels for use aboveground. Delivery, installation and maintenance

Standard	Subject
BS EN 13923:2005	Filament-wound FRP pressure vessels. Materials, design, manufacturing and testing
BS EN 13575:2012	Static thermoplastic tanks for the above ground storage of chemicals. Blow moulded or rotationally moulded polyethylene tanks. Requirements and test methods
BS EN 1778:2000,	Characteristic values for welded thermoplastics constructions. Determination of allowable stresses and moduli for design of thermoplastics equipment
BS EN 12573-1:2000	Welded static non-pressurized thermoplastic tanks. General principles
BS EN 12573-2:2000,	Welded static non-pressurized thermoplastic tanks. Calculation of vertical cylindrical tanks
BS EN 12573-3:2000	Welded static non-pressurized thermoplastic tanks. Design and calculation for single skin rectangular tanks
BS EN 12573-4:2000,	Welded static non-pressurized thermoplastic tanks. Design and calculation of flanged joints
BS EN ISO 16135	Industrial valves. Ball valves of thermoplastic materials
BS EN ISO 16137	Industrial valves. Check valves of thermoplastic materials
BS ISO 9393-1	Thermoplastics valves for industrial applications. Pressure test methods and requirements. General
ISO 9393-2:2005 (R09) Ed 2	Thermoplastics valves for industrial applications - Pressure test methods and requirements. Part 2: Test conditions and basic requirements
BS EN 917:1997, BS 2782-11	Methods 1134A & B Plastics piping systems. Thermoplastics valves. Test methods for resistance to internal pressure and leak tightness
BS EN 1092	Flanges and their Joints – Circular Flanges for Pipes

- (c) Valves, fittings and accessories shall be PN designated.
- (d) Piping, pipe supports, valves, and accessories which are not an integral part of the equipment or are not specified herein are covered in other sections.
- (e) Equipment furnished under this section shall be fabricated and assembled in full conformity with the drawings, specifications, engineering data, test requirements, instructions, and recommendations of the equipment manufacturer unless exceptions are noted by S.O..

- (f) Additional tank general requirements shall include but not be limited to the following items:
 - (i) Nozzle schedule, orientations and elevations as specified in the Particular Specification as required by the various tanks applications;
 - (ii) Tanks shall conform to the appropriate construction and material jointing standards as specified herein;
 - (iii) Test certificates and shop drawings shall be provided;
 - (iv) Tank construction, resin thickness, layers of different materials and materials description shall be as specified in the Particular Specification as much as necessary;
 - (v) Unless otherwise specified, tanks shall undergo an 82 °C 4 hour dry oven post cure after fabrication as per manufacturer's recommendation;
 - (vi) The tank bottom shall be flat within 1 percent of tank diameter;
 - (vii) All tank nozzles shall be finished internally such that there are no exposed fibres;
 - (viii) Nozzles shall extend at least 150 mm from outside face of tank to face of flange; and
 - (ix) All nozzle flanges shall have a minimum thickness of thirty percent greater than the thickness of BS EN 1092-1 PN 10 flanges.
- (g) Coordination - Where two or more units of the same tank are required, they shall be the product of a single manufacturer; however, all the component parts of the same tank need not be the products of one manufacturer.
- (h) Governing Standards - Except as modified or supplemented herein, materials and construction methods shall conform to the applicable provisions of the following standards:
 - (i) Design and construction of vessels and tanks in reinforced plastics, BS EN 13923 and BS EN 13121;
 - (ii) Design and construction of vessels and tanks in polyethylene or high density polyethylene, BS EN 13575: 2012 and ASTM D1998;
 - (iii) Standard Specification for Contact-Moulded Reinforced Thermosetting Plastic (RTP) Laminates for Corrosion Resistant Equipment, ASTM C582;
 - (iv) Standard Specification for Filament-Wound Glass-Fibre Reinforced Thermoset Resin Corrosion-Resistant Tanks, ASTM D3299; and
 - (v) Standard Specification for Contact-Moulded Glass-Fibre Reinforced Thermoset Resin Corrosion-Resistant Tanks, ASTM D4097.
- (i) Nameplates: Each tank shall be provided with a nameplate. The nameplates shall be of stainless steel or white phenolic material with black engraved lettering 75 mm high and shall be mounted on the tank straight shell. The chemical name and the tank tag number shall be engraved on the nameplate.

- (j) Certification Plates: A stainless steel certification plate shall be mounted below each storage tank's nameplate. The following data shall be included on the certification plate:
 - (i) Name of tank fabricator;
 - (ii) Date of manufacture;
 - (iii) Manufacturer's serial number;
 - (iv) Resin designation for entire tank (structural and corrosion barrier); and
 - (v) Maximum allowable concentration and temperature of the specified chemical solution that can be stored safely.
- (k) The following design requirements apply:
 - (i) The design of tanks, including resin selection (unless otherwise specified), wall thickness, methods and locations of support and stiffener requirements shall be prepared and sealed by a designer meeting the requirements of the manufacturer's Quality Assurance Manual.
 - (ii) Coordinate with the manufacturer of the concerned mixers and other equipment to ensure that tank baffle design, mounting nozzle and support design are suitable and of adequate strength for use with the actual mixers and other equipment to be mounted on the tank. Secure the following:
 - 1) Manufacturer drawings of required mixer and other equipment supports;
 - 2) Weight of the mixer or equipment when operating and when not operating;
 - 3) Dynamic thrust forces generated by the mixer or equipment during operation;
 - 4) Description of tank modifications required by mixer and other equipment manufacturer to optimize mixing; and
 - 5) Manufacturer's instructions on assembly, installation and dismantling of the mixer and other equipment.
- (l) Submittals
 - (i) Action Submittals
 - 1) Shop Drawings:
 - Fabricator's catalogue information, descriptive literature, specifications, maximum tensile and compressive stress of the tank material, identification of materials of construction, laminate layers, corrosion barrier, UV resistant additive, fire-retardant additive and complete resin system information;
 - Letter from resin manufacturer stating that selected resin is suitable for intended service;
 - Detailed fabrication drawings;
 - Tank data indicating equipment number, pressure rating, temperature rating, inside and outside diameter, straight shell lengths, overall lengths, wall thickness, empty weight, weight when full, corrosion barrier thickness and details of nozzle design;

- Tank capacity chart indicating litres for each mm of depth and cumulative total from bottom;
 - Fabricator's detailed requirements for tank foundations;
 - Recommended bolt torque for bolted FRP and other thermoplastic connections; and
 - Anchorage and bracing drawings and cut sheets.
- 2) Samples: Laminate sample representative of production quality of surface finish and visual imperfections.
- (ii) Informational Submittals
- 1) Complete design calculations for tanks, supports and appropriate accessories;
 - 2) Anchorage and bracing calculations;
 - 3) Fabricator's Certificate of Compliance with fabrication requirements;
 - 4) Qualifications of Fabricator's Quality Assurance Supervisor;
 - 5) Copy of fabricator's Quality Assurance Program;
 - 6) Quality Assurance Inspection:
 - Qualifications of Independent FRP Quality Assurance Inspector;
 - Initial QA Inspection Report; and
 - Certification of Factory Testing.
 - 7) Certification that mixer and other equipment supports, tank baffle design, and access nozzles have been coordinated with the actual equipment being furnished;
 - 8) Special shipping, storage, protection and handling instructions;
 - 9) Fabricator's printed installation and tank support instructions; and
 - 10) Manufacturer's Certificate of Proper Installation in accordance with the General Specification on Manufacturers' Services.
- (iii) Contract Closeout Submittals: Service records for repairs performed during construction Quality Assurance.
- (iv) Fabricator's Quality Assurance Supervisor: Minimum of 5 years' experience in fabrication of fibreglass structures.
- (v) Structural Designer endorsement by Professional Engineer registered with the Professional Engineers Board, Singapore in appropriate discipline.
- (vi) Independent FRP Quality Assurance Inspection
- 1) FRP inspector with minimum 5 years' experience;
 - 2) A corporately and financially independent organization that can function as an unbiased inspection authority; and
 - 3) Professionally independent of manufacturers, suppliers and installers of systems being inspected.

- (vii) Manufacturer's Field Services write-out elaborating on Manufacturer's Field Services to be submitted for S.O.'s approval.
- (m) Delivery, Storage and Handling
 - (i) Prepare and protect tank for shipment as follows:
 - 1) Mount tank on padded cradles if shipped horizontally or on a suitable skid if shipped vertically;
 - 2) Protect flanged nozzles with wooden blinds bolted to flange and having a diameter of 50 mm greater than outside diameter of flange;
 - 3) Provide either rigid plugs inside end to prevent deflection or wooden boxes for unflanged components. Brace open end of tank with suitable stiffening member to prevent deflection;
 - 4) Do not ship components or other pieces loose inside tank;
 - 5) Load tank with at least 50 mm of clearance between tank (including fittings) and bulkheads, or bed of vehicle; and
 - 6) Regardless of mode of transportation, firmly fasten and pad components to prevent shifting of load or flexing of components while in transit.
- (n) Sequencing and Scheduling
 - (i) The tanks shall not be shipped from the Factory until the S.O.'s review of Certification of Factory Testing is completed and the S.O.'s approval is given.

G4.61.2 Products

- (a) Performance and Design Requirements
 - (i) Each tank shall be designed for storage of the chemical or chemicals at the design conditions specified in the Particular Specification.
 - (ii) FRP and thermoplastic tanks designed for odour scrubber vessels shall refer to the General Specification on Odour Control Systems for details of the specific performance and design requirements. In addition, the Particular Specification shall also be referred to for other application specific requirements.
 - (iii) Usable capacity shall be measured from the invert of the tank overflow nozzle to the top of the pump suction nozzle.
 - (iv) Vertical tanks shall be vertical cylinders with a flat bottom and domed or conical top.
 - (v) Tanks shall be provided with a suitable overflow connection.
 - (vi) Tanks shall be vented and will normally be used to store the specified chemical at atmospheric pressure, but shall be designed to withstand the hydrostatic head resulting from the tank being surcharged to 150 mm above the top of the tank.
 - (vii) Unless otherwise indicated or specified, all tanks will be installed on concrete bases and cushioned using an EPDM sheet of not less than 16 mm thick. The cushion sheet shall have length and width at least 150 mm greater than the tank diameter underneath the tank. The tanks shall be anchored to the concrete base with suitable anchor bolts.

(b) Materials for FRP Tanks

FRP Tanks shall generally be contact moulded according to ASTM D4097, Grade 1.

(i) Resin

- 1) Suitable for intended service;
- 2) Premium grade and corrosion resistant, such as chlorendic polyester, vinyl ester, or bisphenol A fumarate polyester;
- 3) Use same resin throughout entire tank shell;
- 4) Add ultraviolet absorbers to surfacing resin to improve weather resistance;
- 5) No dyes, pigments, or colorants, except in exterior gel coat;
- 6) No fillers or thixotropic agents;
- 7) Curing System:
 - As recommended by resin manufacturer or as specified herein;
 - Cure products as specified in ASTM D4097; and
 - Measure Barcol hardness according to ASTM D2583.
- 8) Post-cure tank and appurtenances in accordance with resin manufacturer's recommendation for duration and temperature. Post-curing shall be completed with warm-to-hot dry air, free of combustion products. Hot spots shall be avoided; and
- 9) For hypochlorite service, no MEKP (Methy Ethyl Ketone Peroxide) cobalt catalyst system shall be allowed in liner. Cure liner with benzoyl peroxide-dimethyl aniline. Structural layer may be cured with either catalyst system.

(ii) Reinforcement

- 1) Surfacing Veil: Chemical surfacing mat, two layers of polyester fabric, 300 to 400 microns thick, with a finish and a binder compatible with the lay-up resin.

(iii) Plastic Laminate

- 1) Consists of inner surface (corrosion barrier), interior layer, and exterior layer (structural layer);
- 2) Meet visual acceptance criteria in ASTM C582;
- 3) Meet requirements of mechanical properties in ASTM D4097;
- 4) Reinforce inner surface with resin-rich surfacing veil as specified herein; and
- 5) After inspection of laminate has been completed, apply an exterior coat in accordance with the requirements of the General Specification on Painting and Protection and/or as per specified in the Particular Specification.

- (iv) Marking
 - 1) Identify each tank with fabricator's name, capacity in litres, design temperature, design pressure/vacuum, specific gravity, pH, resin, minimum thickness, tank number, tank name, and date of manufacture; and
 - 2) Provide permanent marking. Seal decals, labels, etc., into laminate exterior with clear resin.
- (v) External Coating
 - 1) External coating with UV inhibitor.
- (vi) Nozzles (if applicable)
 - 1) Gusset 100 mm or smaller nozzles with conical or plate type gussets. Larger nozzles shall be gusseted, if noted.
 - 2) Finish flush with inside surface of tank, unless otherwise indicated.
 - 3) Gaskets:
 - Provide two per nozzle, 3 mm-thick, full-face elastomeric material having a hardness of Shore A60 plus or minus 5; and
 - Material shall be suitable for intended service.
 - 4) Flanged Nozzles: Per BS EN 1092 PN10.
- (c) Construction
 - (i) The tanks shall be of hand lay-up, spray-up, or filament-wound construction in accordance with the applicable governing standard.
 - (ii) The finished laminate shall be constructed of a single generic type of thermoset resin and shall not contain colorants, dyes, fillers, or pigments unless otherwise specified.
 - (iii) Ultraviolet absorber shall be added to the resin used in the fabrication of tanks for installation in outdoor locations.
 - (iv) The inner surface layer of the tanks shall consist of a resin rich corrosion barrier with a minimum thickness of 120 mils, unless otherwise specified.
 - (v) The surface of the corrosion barrier exposed directly to the corrosive chemical shall consist of a two-ply surfacing veil with a minimum thickness of 20 mils.
 - (vi) The remainder of the corrosion barrier shall consist of two layers or more of chopped strand mat or equivalent.
 - (vii) The inner surface layer of all other tanks shall consist of a resin rich corrosion barrier with a minimum thickness of 110 mils.
 - (viii) Unless otherwise specified, tanks shall be catalysed with a benzoic peroxide/dimethyl aniline (BPO/DMA) catalyst system, and shall undergo an 82 °C 4 hour dry oven post-cure after fabrication as per the resin manufacturer's recommendation.
 - (ix) Use of thixotropic resin additives will not be allowed.

- (x) Alternative catalyst systems will be considered if written confirmation from the resin manufacturer verifying the effectiveness of the proposed system is provided and the S.O.'s approval is obtained.
 - (xi) The top of the tank shall be reinforced as specified in the applicable governing standard.
 - (xii) Additional reinforcement shall be provided as necessary to support the required accessories and personnel loads.
 - (xiii) Tank roofs shall be provided with a non-slip finish over the entire roof surface as recommended by the tank manufacturer.
 - (xiv) The tank bottom shall be flat within 1 percent of tank diameter.
 - (xv) Bracketed flat surfaces shall be provided on the tank for installation of nameplate and certification plate.
 - (xvi) The tank shall be provided with the appropriate number and size of lifting lugs for handling and installation and hold-down lugs for anchoring the tank to the concrete base.
- (d) Materials for PE, HDPE and PP Tanks
- PE, HDPE or PP Tanks shall generally be according to ASTM D1998-13 and BS EN 12573-1 to 4.
- (e) Accessories and Appurtenances
- (i) Supports
 - 1) Tank Supports:
 - Provide for all tanks as shown in the drawing; and
 - Provide a compressible and corrosion resistant sheet of EPDM material (or other alternative material subject to S.O.'s approval) of at least 16 mm thick having length and width of at least 150 mm larger than the tank's diameter underneath each tank to act as a cushion to even out plinth surface and tank base unevenness.
 - 2) Pipe Supports:
 - Provide for all tank overflow pipes, loading pipes, vent pipes, drain pipes and recirculation pipes;
 - Spacing of supports shall be as recommended by fabricator, but shall not be greater than 1200 mm on centre;
 - As shown on Drawings and shall allow removal of pipe; and
 - FRP complete with necessary bolts, nuts, and washers.
 - 3) Mixer and Other Equipment Supports and Connections:
 - Use mixer support design and location requirement obtained from mixer manufacturer for design, fabrication and drilling mounting holes in channels and information on mixer support height.

- Minimum Support System:
 - Two FRP or SS316 channels for each mixer with adequate cross bracing and foot platform for mounting mixer directly on channels;
 - Size support channels for dead and live loads imposed by mixer on the channels and grating, plus 750 kg/m² live load on grating; and
 - Support mixer without vibration.
- (ii) Special Tools and Accessories
 - 1) Equipment requiring special devices for lifting or handling shall be furnished complete with those devices.
- (iii) Access Manways
 - 1) Access manways shall have a nominal diameter of 600 mm. The number and location of access manways on the tank shall be submitted to the S.O. for approval unless otherwise specified.
 - 2) Each manway shall be flanged, fully gasketed and furnished with a fabricated blind flange having the same properties as the tank wall laminate.
 - 3) Flange diameter and drilling shall conform to BS EN 1092-1 PN 10. Side access manways shall be centred approximately 900 mm above the bottom of the tank.
- (iv) Flanged Nozzles
 - 1) Nozzles for connecting piping and accessories shall be provided on each tank. The locations and of the sizes of the nozzles shall be submitted to S.O. for approval unless otherwise specified
 - 2) Each nozzle shall be flanged with flange diameter and drilling conforming to BS EN 1092-1 PN 10.
 - 3) Nozzles shall extend at least 150 mm from outside face of tank to face of flange.
 - 4) Flanged nozzles shall be fabricated of the same material as the tank and shall be gusseted to the tank or otherwise reinforced in accordance with the governing standard. All tank nozzles shall be finished internally such that there are no exposed fibres.
 - 5) All nozzle flanges shall have a minimum thickness of thirty percent greater than the thickness of BS EN 1092-1 PN 10 flanges.
 - 6) The mounting flange for ultrasonic type transmitters shall be not less than 450 mm above the maximum liquid level.
 - 7) The centre-line of the nozzle shall be at least 600 mm from the opposite tank sidewall, any other fill nozzle, and other obstructions.
 - 8) Nozzles for drain connections shall be installed so the invert is as low as possible to allow maximum draining of the tank.

(v) Flange nozzle tolerances and design requirements

Flange nozzle tolerances shall follow the values indicated below as per ASME RTP-1-2005.

- 1) Flange Flatness Tolerance: ± 0.8 mm.
- 2) Eccentricity between bolt holes and centre of nozzle: for size 50 mm, ± 0.76 mm; for size 80 mm and larger, ± 1.52 mm.
- 3) Nozzle offset from tank wall or top: Nozzle offset from tank wall or tank top shall have a maximum permissible angular deviation of 1 degree from the nozzle face.
- 4) Bolt hole off centres: For four holes bolt patterns, bolt holes shall align 45 degrees off of a vertical and horizontal orientation. For six holes bolt patterns, bolt holes shall align in the horizontal rather than the vertical orientation.

(vi) Vents

- 1) The tank shall be provided with a vent of the size recommended by the manufacturer, to prevent pressurizing the tank during filling or drawing a vacuum inside the tank during pumping or draining.
- 2) Every vent shall be fitted with an insect screen. For elevated tanks, the mesh size shall be Mesh No. 4 (sieve opening 4.75 mm) whereas for underground tanks and tanks on ground level, the mesh size shall be Mesh No. 24 (sieve opening 0.71 mm).
- 3) The detail of the vent shall be submitted to S.O. for approval unless otherwise specified.

(vii) Platforms, Ladders, Handrails and Kick plates

- 1) Materials of Construction as follows:
 - Tank: PE, HDPE or PP as specified according to the relevant codes and standards;
 - Fasteners: FRP;
 - Supports, FRP: Locate as required for field installation of ladders, platforms, or handrails;
 - Anchor Bolts: Type 316 stainless steel bolts, sized by fabricator, and at least 12 mm diameter, and as specified in the General Specification on Metalwork; and
 - Manway Bolts: Type 316 stainless steel bolts, nuts and washers.
- 2) Lifting Lugs: Provide suitably attached for tank weighing over 40 kg.
- 3) Manhole Covers: Top and side manhole covers.
- 4) Sight level tube on the side of tank indicating level in tank.
- 5) Tanks with a total height greater than 4800 mm from the finished floor shall be provided with a safety cage. The ladder shall be bracketed to the tank, supported on and anchored to the concrete base.

- 6) Where specified as required, safety handrails and kick plates shall meet current OSHA requirements (such as OSHA standards 1910.23, 1910.24, 1910.27, 1910.29 and 1910.30) and shall be provided around the top of each tank. The handrails shall be attached to the ladder side rails and anchored to the tank head. The ladder shall be fabricated of galvanized mild carbon steel completely coated with a two-part epoxy coating as specified herein.
 - 7) Uncoated or exposed carbon steel parts or fasteners will not be acceptable. The ladder shall have a clear width of at least 400 mm, with rungs at least 18 mm in diameter and spaced not more than 300 mm apart, and a clearance of at least 175 mm between the back of the ladder and the tank wall.
 - 8) Tank handrails, ladders and lifting lugs shall be painted yellow to meet Singapore safety standards.
 - 9) Epoxy Coating Protection System for Ladders
 - The epoxy corrosion protection system shall be a two component, 100 percent solids, solvent-free, epoxy resin, silica filled, high-build protective and waterproofing coating.
 - Surface primer shall be an epoxy resin containing conductive fillers, 100 percent solids content. Finish coats shall be a two component, 100 percent solids, epoxy resin.
 - The corrosion protection system shall be applied in three or more coats. The protection system shall consist of one prime coat and two or more finish coats.
 - The prime coat wet film thickness shall be at least 4 mils [100 µm]. Each finish coat shall be 15 to 20 mils [375 to 500 µm] dry film thickness.
 - The protection system shall have a total dry film thickness of at least 40 mils [1000 µm].
- (f) Source Quality Control
- (i) Independent PE/HDPE/PP Tank Inspector:
 - 1) To be present at point of manufacture when fabrication is started to perform the following:
 - Observe manufacturing methods, machinery, and techniques to ensure compliance with industry standards and these specifications;
 - Observe initial fabrication to verify compliance with these specifications;
 - Observe quality control methods for mixing resins and testing of completed equipment;
 - Generally, observe quality of other ongoing fabrication; and
 - Prepare Initial QA Inspection Report.

- 2) To be present at point of manufacture upon completion of fabrication and prior to shipment to perform or witness the following:
 - Visual inspection to requirements of ASTM C582 for FRP material or D1998 – 13 for PE/HDPE material;
 - Barcol Hardness measurements per ASTM D2583;
 - Acetone sensitivity test for internal secondary bonds;
 - Glass content by ignition loss on three cut-outs as per ASTM D2584 for FRP material;
 - Requirements for PE/HDPE tanks material as per BS EN 13575; and
 - Hydrostatic Leak Test:
 - Perform on each tank;
 - Fill to top nozzle; allow to stand for 2 hours with no visible leakage; and
 - Repair work shall only be carried out when deemed acceptable by the Independent PE/HDPE/PP tank Inspector and approval is given by the S.O..
- 3) Identify and retain cut-outs. The S.O. may select certain cut-outs for testing for physical properties of laminate.
- 4) Factory Test Reports: Certify results, by signature, of the following:
 - Inspections;
 - Results of hydrostatic testing; and
 - Test reports of physical properties of standard laminates.

G4.61.3 Execution

- (a) Installation
 - (i) In accordance with fabricator's written instructions;
 - (ii) Accurately place anchor bolts using templates furnished by fabricator, and as specified in the General Specification on Metalwork; and
 - (iii) Interface with Other Products: Coordinate with mixer and other equipment manufacturer as required for proper mounting of tank mixers.
- (b) Field Quality Control
 - (i) Functional Test:
 - 1) Conduct on each tank; and
 - 2) Hydrostatic leak test with tank full of clean water. Allow water to stand for 24 hours to verify no leakage. Contractor shall be responsible to discharge the water to the location approved by the S.O..

- (c) Manufacturer's Field Services
 - (i) Provide fabricator's representative at Site in accordance with the approved Manufacturers' Field Services write-out to carry out all required field services.

END OF SECTION